

1. Expand $x^2y + 3y - 2$ in the power of $x - 1$ and $y + 2$ using Taylor's series.
2. Expand $e^x \cos y$ about $(0, \frac{\pi}{2})$ up to the third term using Taylor's series.
3. Show that the expansion of $\sin(xy)$ in the power of $(x - 1)$ and $(y - \frac{\pi}{2})$ up to and including second degree term is $1 - \frac{1}{8}\pi^2(x - 1)^2 - \frac{1}{2}\pi(x - 1)(y - \frac{\pi}{2}) - \frac{1}{2}(y - \frac{\pi}{2})^2$.
4. A robot moves with very small angles x and y . Its motion is represented by $f(x, y) = \cos x \cos y$. Find the Taylor expansion near $(0, 0)$.
5. A probability model in data science is defined by $f(x, y) = \sqrt{1 + x + y}$. Expand the function about $(0, 0)$ up to third-order terms.
6. Find the maxima and minima of the function $f(x, y) = x^3 + y^3 - 3x - 12y + 20$.
7. Show that the function $(y - x)^4 + (x - 2)^4$ has a minimum value at $(2, 2)$.
8. Show that the function $f(x, y) = x^2 - 3xy^2 + 2y^4$ has neither a maximum nor a minimum value at the origin.
9. A data center adjusts two system parameters x and y to reduce power consumption given by $f(x, y) = x^2 + 2y^2 - 8x - 4y$. Find the minimum power consumption.
10. A game engine optimizes graphics performance using $f(x, y) = x^3 + y^3 - 3xy$. Find the stationary points and test for maxima or minima.
11. Find the smallest and largest value of $2x - y$ on the curve $x - \sin y = 0, 0 \leq y \leq 2\pi$.
12. Find the shortest distance between the line $y = 10 - 2x$ and the ellipse $\frac{x^2}{4} + \frac{y^2}{9} = 1$.
13. Find the extreme values of $f(x, y) = 2x + 3y + z$ such that $x^2 + y^2 = 5$ and $x + z = 1$.
14. A data center wants to minimize energy consumption $f(x, y) = x^2 + y^2$ subject to network capacity $x + 2y = 20$. Apply Lagrange multipliers to find the minimum.
15. A game developer wants to maximize graphics quality $f(x, y) = x^2y$ subject to $x + y = 12$. Find the optimal values using Lagrange multipliers.

Answers:

1. $-10 - 4(x - 1) + 4(y + 2) - 2(x - 1)^2 + 2(x - 1)(y + 2) + (x - 1)^2(y + 2)$
2. $-\left(y - \frac{\pi}{2}\right) - x\left(y - \frac{\pi}{2}\right) - \frac{1}{2}x^2\left(y - \frac{\pi}{2}\right) + \frac{1}{6}\left(y - \frac{\pi}{2}\right)^3 + \dots$
4. $f(x, y) = 1 - \frac{1}{2}(x^2 + y^2) + \frac{1}{24}(x^4 + 6x^2y^2 + y^4) + \dots$
5. $f(x, y) = 1 + \frac{x+y}{2} - \frac{1}{2}(x^2 + 2xy + y^2) + \frac{1}{16}(x^3 + 3x^2y + 3xy^2 + y^3) + \dots$
6. Local minima at (1,2) and local maxima at (-1,-2).
9. Minimum power consumption occurs at $(x, y) = (4, 1)$ and minimum value is -18 .
10. Stationary points are $(0, 0), (1, 1)$.
11. Smallest value $= -\sqrt{3} - \frac{5\pi}{3}$ and largest value $= \sqrt{3} - \frac{\pi}{3}$.
12. $\sqrt{5}$.
13. Maximum value is $1 + 5\sqrt{2}$ and minimum value is $1 - 5\sqrt{2}$.
14. Minimum energy consumption is 40.
15. The optimal values that maximize the graphics quality function are $(x = 8)$ and $(y = 4)$, yielding a maximum quality value of $f(8, 4) = 256$.